INF221 - [Modern] Web Design & Development

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Learning Outcomes

The aim of this lecture is to:

- ❖ Introduce you (students) to the Web Design and Development module, and discuss students' commitments and general expectations
- ❖ Provide students with an understanding of web design and development general concepts and principles which guided the growth of web technology as we it today
    - ➢ Discuss genesis and the nature of the Web: the World Wide Web (WWW)
    - ➢ Discuss challenges and benefits of the Web as a tool for effective communication and information sharing on the Internet
- ❖ Introduce you (students) to **modern** tools, methods, standards and technologies in *website* and *web-application* **design and development**.

Indicative Content

- ❖ General Expectations
- ❖ The nature of the Web
- ❖ Challenges
- ❖ Benefits or Advantages
- ❖ Tooling
- ❖ Soft skills
- ❖ Summary
- ❖ References

Frameworks & Libraries

Languages

Databases & Servers

Others

But where do we start from?

An Ideal Web application architect or Software Engineer

Only through a fundamental understanding of the core technology can developers or engineers be expected to grow with the rapid technological changes associated with Web application development.

- Must be ***"the jack of all trades and master of all"*** *as opposed to* ***"the jack of all trades yet master of none"***

The Nature of the Web

There are 3 essential components which defines the web technology – Tim Berners-Lee (1990). He also wrote the first web editor and server - *httpd* (Google up).

- MarkUp
  - HTML: HyperText Markup Language.
  - The markup (structure) language for formatting & cross-referencing hypertext document via hypertext links on the web
- Protocol
  - HTTP: Hypertext Transfer Protocol.
  - Allows for the retrieval of linked resources from across the web.
  - Protocol for transporting specialized messages over a network - core foundation of the WWW
- Uniform resource access notation
  - URI: Uniform Resource Identifier.
    - A kind of “address” that is unique and used to identify to each resource on the web. It is also commonly referred to as a **URL**.
  - Uniform resource notation for addressing accessible resources over the network (URI – Uniform Resource Identifier or URL – Unified Resource Locator)
    - Generalized notation - **URLs**
    - scheme://host[:port#]/path/.../[;url-params][?query-string][#anchor] – *to be explored later on*

The Nature of the Web…

Therefore, the WWW is the **universe of network-accessible resources**, which may be interlinked by **hypertext** (connection of documents and information resources through links)

- and are **accessible** on the Internet using the *Hypertext Transfer Protocol* (**HTTP**), & other **communication protocols**
  - “The World Wide Web (known as "WWW', "Web" or "W3") is the universe of network-accessible information”
  - an **information space** where documents and other **web resources** are identified by **Uniform Resource Locators** (URLs, i.e. https://www.example.com/)
    - which may be interlinked by hypertext, and accessible via the Internet”
- “The WWW is combination of all users and resources on the Internet that are accessible via the Hypertext Transfer Protocol (HTTP).”

The Web - Key Initial Guiding Principles

- Decentralisation
  - No central authority to control creation and access of documents
  - No single point of failure
  - Implying freedom from indiscriminate censorship and surveillance
- Non-discrimination
  - Similar quality of communication by all parties regardless of what tools everyone is using to connect
  - Principle of equity is also known as Net Neutrality.
- Universality
  - Communication of any two computers must be based of a common language
    - no matter what different hardware people are using;
    - where they live; or
    - what cultural and political beliefs they have.
  - Allows diversity to flourish
- Consensus
  - For universal standards to work, everyone had to agree to use them
  - Tim and others achieved this consensus by giving everyone a say in creating the standards, through a transparent, participatory process at W3C.

Genesis of the Web

- The **Web**, as we know it today, can *partially* be attributed to its grass roots proliferation as a **tool for personal publishing**
- The technology was simple:
  - A **computer** connected to the **Internet**,
  - Running a **Web server**,
  - Serving **stored** documents or files to **Web clients (computers too)**
- Initial technology was quite simple in nature
  - During its infancy, web sites served static content - found in files/documents
  - Very few information services were truly considered dynamic
- Dynamic information services:
  - Services that connected the Web to relational databases enabled the web to become truly dynamic.
  - No longer was it sufficient to say that you were designing a ‘Website’ (as opposed to a mostly collection of ‘Web pages’)
  - It became necessary to designing & developing Web applications

Categories - **Web Products**

Web products may take many forms, but can be categorised into two main categories

- ❖ Website
- ❖ Web application

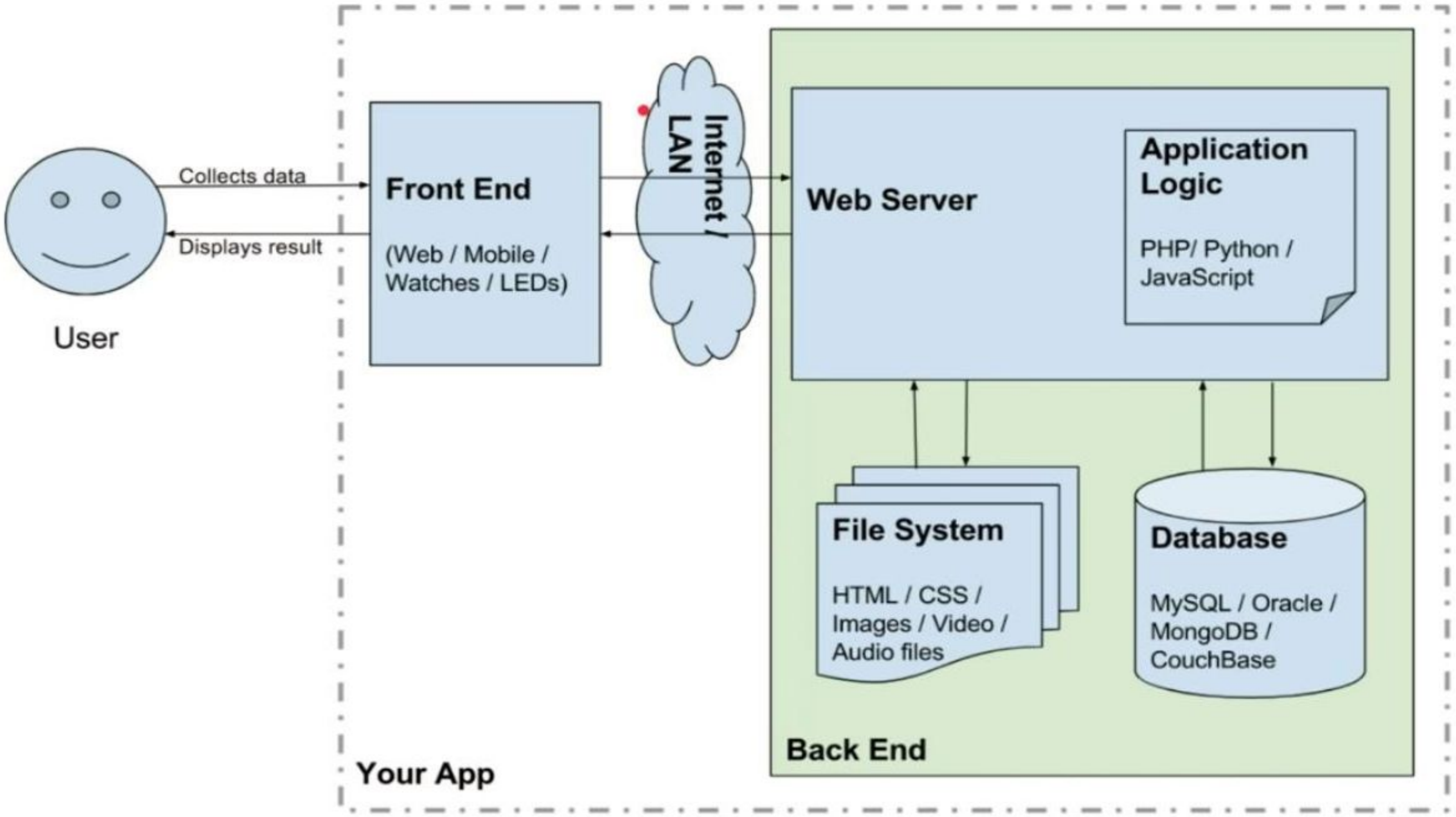
The difference is in how each category can be developed

- ❖ organization or
- ❖ architectural attributes or
- ❖ nature of the resulting artefact - product or
- ❖ in each instance, the nature of what is returned as response when the product is run in the client application

Categories - **Web Products**

- Web site
  - A website is a collection of related web resources, such as web pages, multimedia content, which are typically identified with a common domain name, and published on at least one web server
  - Defined by their content and information
  - A Web site simply delivers content from static files
- Web application
  - A web application or web app is a **dynamic** software/computer program defined by user **interactions (driven by user actions)** running on **client computers** in either **web browsers or web servers** (Examples???)
  - It presents dynamically tailored content based on request parameters, tracked user behaviors, and security considerations

By definition, a web application is a **client/server application** that uses a **Web browser** as its client program, and performs an **interactive service by connecting with servers** over the Internet (or Intranet) **to generate dynamic web pages** based on request **parameters**, tracked/stored **user behaviors**, and **security** considerations


User
Collects data
Displays result
Front End
(Web / Mobile / Watches / LEDs)
Internet / LAN
Web Server
Application Logic
PHP/ Python / JavaScript
File System
HTML / CSS / Images / Video / Audio files
Database
MySQL / Oracle / MongoDB / CouchBase
Back End
Your App

Definitions - **Design** vs **Development**

Web **design** and **development** are broad concepts. Which require breaking down into key sub-concepts

- Web design
  - A process or activity which involves the application of various disciplines and skill sets for the production or maintenance of websites or web applications
  - Web design encompasses many different skills and disciplines in the production and maintenance of websites or applications. These include but not limited to:
    - Graphic design; Software Programmers or Developers;
    - User Interface (UI) design; User Experience (UX) design;
    - Search engine optimization (SEO);
- Web development
  - Set of activities involved in the development of a website or web application for the Internet (World Wide Web) or an intranet (a private network). May include several activities such as planning - ideation and discovery, design, testing, maintenance and support.
  - Actual work process activities involved in the production of a website or web application
    - not limited to web design

Key Components of a Web Application

- ❖ Webpage - Set of User Interface Elements
  - ➢ Structured and designed using a markup language, presentation styles and scripting languages
- ❖ Web Server
  - ➢ For hosting a collection of web pages
- ❖ Database
  - ➢ Storage of data and any user generated information

Take note that in this module you are required to have glean essential understanding of all these components

- ❖ Individual purpose in any singular application
- ❖ And how they work together in any particular web application?

Challenges - Web Design & Development

- Collaboration
  - Professional web development is not an individual task
    - Development team members may reside in different time zones and geographical locations
    - Project management may proof a complex and difficult activity
  - Thus, may require the use of collaborative tools like GIT and its services
    - such as GitHub, GitLab, BitBucket etc
- Technological Complexity
  - Web development involves navigating a vast landscape of programming languages, frameworks, libraries, and tools, each with its own learning curve and best practices
- Browser Compatibility and Accessibility
  - Ensuring that websites function correctly across different web browsers and devices can be challenging
    - Their design must take into consideration variations in rendering engines, standards support, and user preferences
- Performance Optimization
  - Optimizing website performance for speed, responsiveness, and scalability requires careful consideration of factors such as
    - Code efficiency, server configuration, caching strategies, and content delivery networks (CDNs).
- Security Concerns
  - Websites are vulnerable to various security threats, including malware, data breaches, cross-site scripting (XSS), and SQL injection attacks.
  - Web developers must implement robust security measures, such as:
    - encryption, authentication, and access controls,
  - These are essential in protecting sensitive information and maintain user trust.
- Maintenance, Support and Updates
  - Websites require ongoing maintenance to ensure they remain functional, secure, and up-to-date with evolving technologies and user expectations.
  - This includes performing regular backups, applying security patches, fixing bugs, and adding new features as needed.

Benefits - Web Design & Development

- E-commerce
  - enables business users to engage in online business transactions
  - facilitates market expansion by reaching traditional brick-and-mortar locations through ***digital*** store fronts and channels
- Accessibility
  - Efficient storage and delivery of information
    - provides an accessible platform for information and services, allowing people to connect, communicate, and conduct transactions regardless of geographic location or physical limitations
- Communication
  - Websites serve as a vital communication tool, allowing organizations to share news, updates, and resources with their audience,
  - as well as gather feedback through interactive features like contact forms and comment sections.
- Marketing
  - Websites play a crucial role in digital marketing strategies
    - providing a platform for
      - content marketing and monetisation,
      - search engine optimization (SEO),
      - social media integration,
      - and online advertising campaigns.
  - Enhances online visibility, engagement

Tooling - Web Design & Development

- ❖ Project Planning and Management
- ❖ Collaboration and Software Development
- ❖ Design
- ❖ Testing
- ❖ Deployment

Soft Skills [**MUST HAVEs**]

The web constantly evolves and technologies and trends are always changing. As a **UNIVERSITY** student you need to constantly update your knowledge to keep up.

- Constant learning mindset
  - Subscribe to key information outlets on the web ie **technical news**, **blogs**, and release notes to key technologies like **languages**, **frameworks.**
    - Recommendations: **medium**, **ByteByteGo, Codecademy, freeCodeCamp, MDM** etc
  - Engage in reading tasks or small research projects regularly and periodically
  - Be curious and set aside specific times for acquiring new knowledge and skills. Be deliberate.
- Open to embracing failure
  - The fear of failure limits many new developers and students alike from experimentation and risk taking
  - What is the value of failure? What lessons can you learn now that can be relevant in future similar situations?
    - **Lessons from the past**, should determine how we live in the present life and future possibilities.
  - Tips for improvement?
    - Peers,
    - Look to Communities and group meetups for moral and technical support,
    - Get feedback from your lecturers/instructors/tutors on personal or academic assignments.

Soft Skills [**MUST HAVEs**]...

- ~~Constant learning mindset~~
- ~~Open to embracing failure~~
- Effective research(er)
  - Spend a good amount of time searching for relevant information and possible solutions to web project problems.
    - Do not pester your peers or colleagues to help you every time you meet a roadblock.
  - Strategies to effectively conduct your research:
    - Consulting the documentation
    - Search engines - Google Scholar and others
    - Make effective use of an **experienced contact's** time.
    - Choosing valid information sources
      - look for modern approaches to problem solving, some are obsolete…consider not using them
    - Using social media effectively - benefit from professional networks and contribute to professional communities
    - Cautiously use AI - familiarise yourself with its strengths and weaknesses
      - understand that this is an academic institution where you are required to to do research and generate knowledge for the good of the nation and communities you are coming from.
- Collaboration and teamwork
  - Professional web development is not an individual task - learn to work in teams, conduct code reviews. Learn to be humility, conflict resolution, and cooperation.
  - Learn to be an effective communicator. Feedback must be constructive and respectful.
  - Join a Slack channel, Discord, or a similar space, ask peers for help, share resources.

Soft Skills [**MUST HAVEs**]...

- ~~Constant learning mindset~~
- ~~Open to embracing failure~~
- ~~Effective research~~
- ~~Collaboration and teamwork~~
- Build relevant skills today for tomorrow’s job interview
- Relevant contextual information
  - Understand why things happen they way they do.
  - Understand where to use what concept and how…:
    - loops - understand **how** and **when** to use then,
    - objects (OOP - javascript) - their relevance in web system modelling, design and development
    - functions (javascript/typescript)

Soft Skills [**MUST HAVEs**]...

- ~~Constant learning mindset~~
- ~~Open to embracing failure~~
- ~~Effective research~~
- ~~Collaboration and teamwork~~
- ~~Build relevant skills today for tomorrow's job interview~~
- ~~Relevant contextual information~~
- Workflows and processes
  - Understand a bigger picture about a web project - having knowledge about a specific language alone is not enough.
    - Learn about various frameworks, tools, systems, and job roles that go together to deliver an entire web application
    - Understand the following at a high level:
      - technology combinations and application architectures
      - processes for a technical project, including where different tools are used in those processes
      - job roles, and where they are involved in those processes
      - Common project management models and methodologies, such as agile and waterfall etc.

Summary

References

- History of the Web - World Wide Web Foundation
- https://github.com/mdn/curriculum/blob/main/curriculum/1-getting-started/1-soft-skills.md
- https://github.com/mdn/curriculum/blob/main/curriculum/1-getting-started/2-environment-setup.md